Diabetes Risk Prediction with Multi-Layer Perceptrons

Results and interpretation report

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Companion file: *diabetes_risk_mlp.ipynb* (all cells executed; every number in this report is taken from that run).

Summary

We built classifiers that predict a diabetes diagnosis (`diagnosed_diabetes`) from 100,000 *synthetic* patient records using **only multi-layer perceptrons**. Because the data contain blood tests that almost give the answer away, two situations are studied side by side: **S1 - with lab tests** (diagnosis support) and **S2 - without lab tests** (screening from demographics, lifestyle, vitals and lipids). We evaluated 240 configurations (432 training runs): which columns to use, preprocessing settings (encoding, scaling, clipping, engineered features, PCA, correlation filter), 22 architectures, 7 activations, regularisation, optimisers, batch sizes, schedules, loss functions and a 30-trial random search per scenario, plus a scikit-learn cross-check.

	S1 - with lab tests	S2 - without lab tests
Final MLP	4 x 16 ReLU, dropout 0.2 (1,585 parameters), ensemble of 5	4 x 16 ELU, AdamW (1,521 parameters), ensemble of 5
Test accuracy (95 % CI)	**92.0 %** (91.6 % - 92.5 %)	**62.8 %** (62.0 % - 63.5 %)
Test ROC-AUC (95 % CI)	**0.944** (0.941 - 0.948)	**0.652** (0.643 - 0.660)
Sensitivity / specificity	0.869 / 0.998	0.810 / 0.354
Majority-class baseline	60.0 %	60.0 %

Main findings. (1) With lab tests the MLP is accurate (92 %), but it essentially **re-discovers the clinical rule HbA1c >= 6.5** (91.0 % accuracy on its own): HbA1c is by far the most important input. (2) Without lab tests only a weak signal exists (AUC 0.65; 62.8 % accuracy vs 60.0 % for always guessing "diabetic"), carried by family history, age and physical activity. (3) The dataset contains a **target leak** - `diabetes_stage` gives 99.9 % accuracy - which must be excluded. (4) Unlike the image task, these networks do not over-fit; extensive tuning of architecture, regularisation and optimiser changes the AUC by less than 0.002, so the limit is the information in the data, not the model.

1. Task, data and protocol

Task and constraint. Predict whether a patient has diagnosed diabetes (0 / 1) with MLPs only. PyTorch is used as a fast engine for fully-connected layers; a "0-hidden-layer network" (logistic regression) is evaluated only as a reference and is never selected, because it is not an MLP. scikit-learn's `MLPClassifier` serves as an independent cross-check. Scalers, PCA, etc. are preprocessing for the MLP.

Data. The file `diabetes_dataset.csv` has 100,000 rows and **31 columns** (the Kaggle description says "35+"), 24 numeric and 7 categorical, with **no missing values and no duplicate rows**. **60.0 % of the patients are diagnosed** (the description claims 20-25 %). The data are **synthetic** (generated from statistical distributions), so conclusions cannot be transferred to real patients.

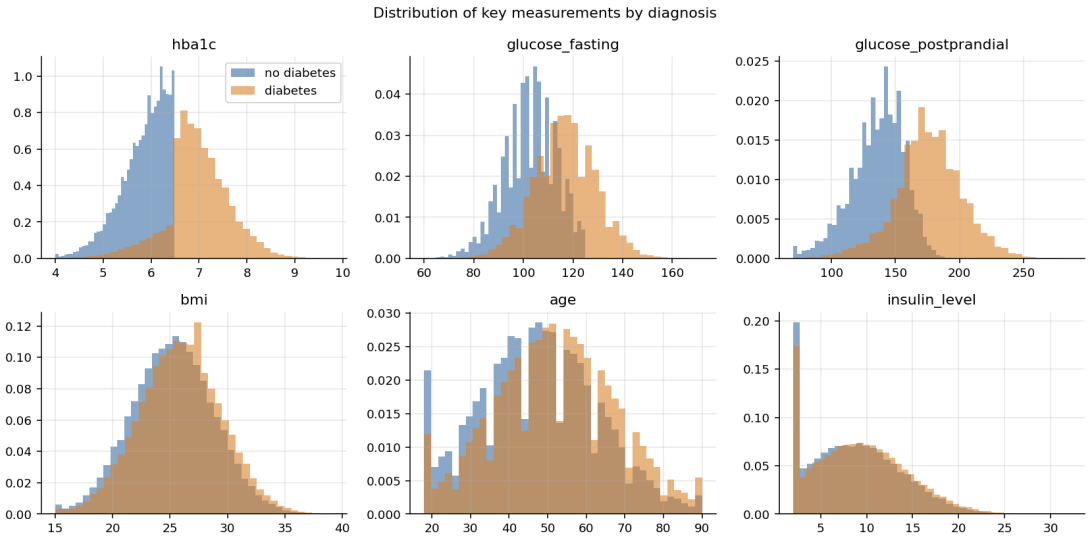


Figure 1. Distributions of key measurements by diagnosis. HbA1c and the two glucose values separate the classes strongly; BMI, age and insulin hardly do.

Target leakage. `diabetes_stage` is defined from the diagnosis: all 59,774 "Type 2" patients are positive, all 39,826 "No Diabetes" / "Pre-Diabetes" patients are negative, and only the 400 Gestational / Type 1 rows are mixed. A model given this column would reach 99.9 % accuracy without learning anything about risk (Section 3.2), so it is excluded from every model. `diabetes_risk_score` is a pre-computed score that is almost a linear function of the other columns (R^2 = 0.993); it is excluded from the main models and tested separately.

diabetes_stage	diagnosed = 0	diagnosed = 1
Type 2	0	59,774
Pre-Diabetes	31,845	0
No Diabetes	7,981	0
Type 1	56	66
Gestational	120	158

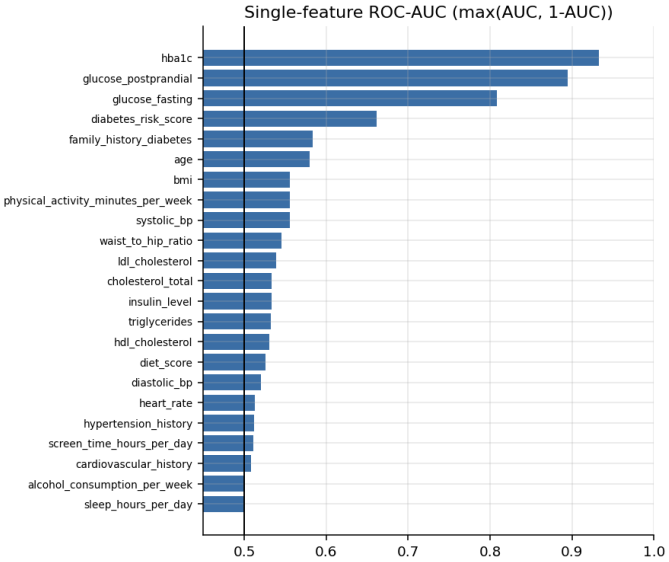


Figure 2. Single-feature ROC-AUC. HbA1c alone reaches 0.932, post-meal glucose 0.894, fasting glucose 0.808; every other input column is below 0.59 (family history 0.584, age 0.580); the pre-computed risk score reaches 0.661 but is excluded. The clinical rule HbA1c >= 6.5 alone gives 91.1 % accuracy on the whole data.

Scenarios and split. S1 uses all 28 remaining columns (including fasting / post-meal glucose, insulin and HbA1c); **S2** uses the 24 columns without these four laboratory values. Stratified 70 / 15 / 15 split (70,000 train, 15,000 validation, 15,000 test). All statistics (scalers, PCA) are fitted on the training rows only; **all design decisions use the validation set (ROC-AUC) and the test set is used once for the final models**. Accuracy,

F1, sensitivity, specificity, Brier score and calibration are also reported.

2. Models and preprocessing

MLP family. `Linear -> [BatchNorm] -> activation -> [Dropout]` blocks and one output logit (sigmoid = probability of diabetes), trained with binary cross-entropy (or focal loss), early stopping on validation AUC (patience 8, at most 40 epochs). Training is fast (a run takes 1-10 s on the GPU), so 432 runs took 42 minutes. Every finished run is checkpointed on disk, so an interrupted notebook can resume and returns exactly the numbers of the run that produced them.

Preprocessing options (each fitted on training data only): categorical encoding (one-hot / ordinal for education and income / integer codes); scaler (standard, min-max, robust, quantile-normal, power, none); winsorising at the 1st / 99th percentile; engineered features (pulse pressure, cholesterol / HDL ratios, non-HDL, TG / HDL, glucose delta, HOMA-IR, age x BMI, log transforms); PCA keeping 95 % of the variance; dropping one column of every pair with |r| > 0.9.

Tuning protocol. One factor at a time around the current best configuration (two seeds); a value is adopted only if it improves validation AUC by more than max(0.002, seed std); the adopted values are combined and re-checked. A 30-trial random search then explores interactions, and the five best distinct configurations seen anywhere are re-trained with three fresh seeds before the final choice.

3. Experiments and results (validation set)

3.1 Baseline

The baseline MLP (128-64, ReLU, Adam, batch 512) reaches **AUC 0.942 / accuracy 90.8 %** in S1 and **AUC 0.662 / accuracy 63.3 %** in S2 (3 seeds, std <= 0.0005). Training accuracy (91.1 % / 63.5 %) equals validation accuracy, and early stopping selects epoch 6 and ~3: there is **no over-fitting**; the network learns what there is to learn almost immediately.

3.2 Which columns does the model get? (feature sets and leakage)

Input columns	# inputs	Val. AUC	Val. accuracy
S1	28	0.9421	0.9079
S2	24	0.6621	0.6330
S2+hba1c	25	0.9413	0.9056
S2+glucose (2)	26	0.9127	0.8389
S2+insulin	25	0.6626	0.6329
lab tests only	4	0.9366	0.9149
S1+risk_score	29	0.9429	0.9083
S1+stage (LEAK)	29	1.0000	0.9991

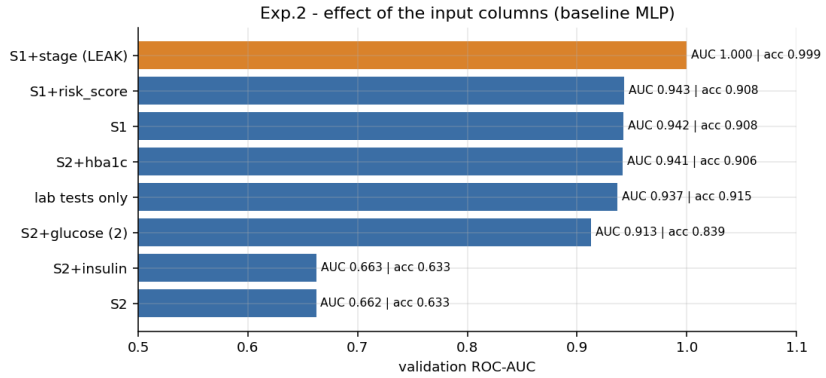


Figure 3. Effect of the input columns on the baseline MLP (orange: leaking column).

Interpretation. The lab tests carry almost all of the signal: adding only HbA1c to the no-lab columns raises AUC from 0.662 to **0.941** (all of S1: 0.942), and the four lab values alone give 0.937 - lifestyle, demographics

and vitals add almost nothing once HbA1c is known. Both glucose values without HbA1c reach only 0.913; insulin adds nothing. `diabetes_risk_score` does not help (0.943 vs 0.942, within noise), as expected for a redundant, derived column. Adding `diabetes_stage` yields AUC 1.000 and 99.9 % accuracy, a perfect score that merely re-states the label: any near-perfect result on this dataset should be checked for this leak.

3.3 Preprocessing

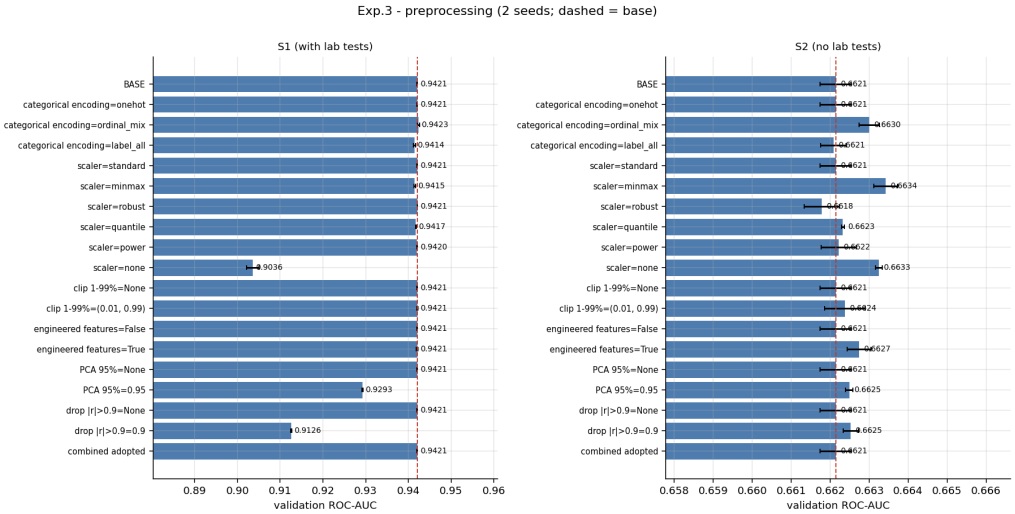


Figure 4. Preprocessing options (two seeds; dashed line: default one-hot + standardisation).

Interpretation. In S1 scaling only matters when it is missing: with raw inputs AUC drops to 0.904 (accuracy 82.9 %) because the columns have very different ranges, while all genuine scalers agree within 0.001. Encoding, winsorising and engineered features (pulse pressure, cholesterol ratios, HOMA-IR) change nothing: an MLP can form such combinations itself, and HbA1c dominates the signal. Two options *hurt*: PCA (0.929) discards low-variance directions that are still predictive, and the correlation filter (0.913) **removes HbA1c** because it correlates 0.93 with post-meal glucose - a correlation filter is unsafe when the redundant-looking column is the best predictor. In S2 all variants lie in 0.662-0.663 (differences are noise). Nothing was adopted in either scenario.

3.4 Architecture

Validation AUC for 1-4 hidden layers x 16 / 64 / 256 / 512 units (two seeds). Linear reference (no hidden layer): S1 0.9287, S2 0.6641.

Hidden layers \ units	16	64	256	512
1	0.9416	0.9417	0.9411	0.9412
2	0.9424	0.9421	0.9424	0.9425
3	0.9423	0.9428	0.9423	0.9428
4	0.9441	0.9430	0.9425	0.9425

Hidden layers \ units	16	64	256	512
1	0.6634	0.6632	0.6633	0.6627
2	0.6631	0.6624	0.6622	0.6620
3	0.6637	0.6616	0.6623	0.6622
4	0.6637	0.6619	0.6624	0.6621

Left: S1 (with lab tests). Right: S2 (no lab tests).

Interpretation. The surface is flat (S1 0.9411-0.9441, S2 0.6616-0.6637) and the best network in both scenarios is the small **4 x 16** model (1,585 parameters); more width or depth cannot add information that is not in the columns. **In S1 hidden layers are necessary**: the linear reference reaches only AUC 0.929 / accuracy 85.4 % vs 0.944 / 91.8 %, because the diagnosis follows a threshold-like rule on HbA1c, which a few ReLU units can represent and a linear model cannot. **In S2 the linear model is as good as any MLP** (0.6641 vs 0.6637): there is no non-linear structure to exploit.

3.5 Activation functions

S1: ReLU 0.9441, GELU 0.9440, tanh 0.9430, SiLU 0.9426, Leaky-ReLU and ELU 0.9424, sigmoid 0.9343. S2: all seven activations lie within 0.6637-0.6641. ReLU-type units suit a step-like rule and sigmoid is the weakest (about one point lower); in S2 the activation is irrelevant.

3.6 Regularisation

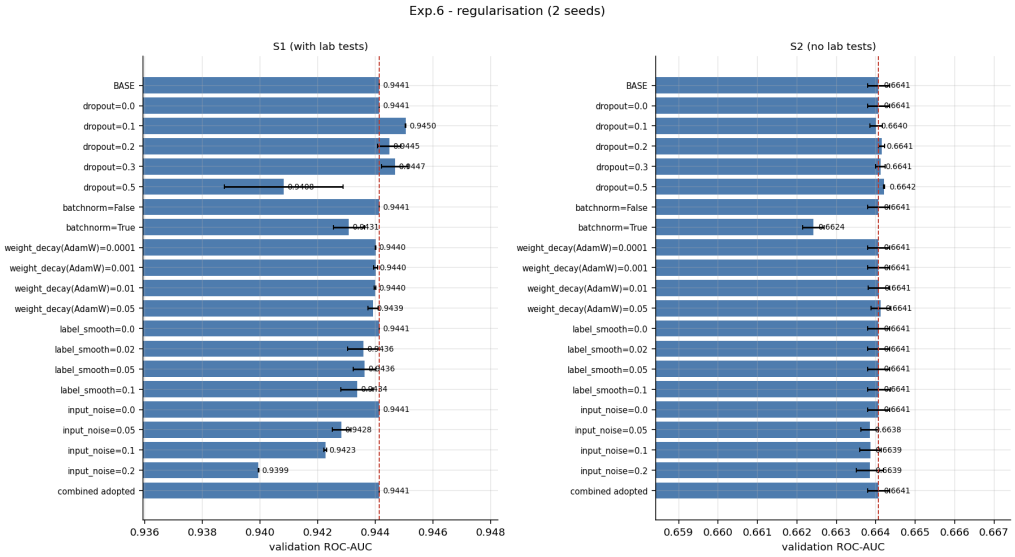


Figure 5. Regularisation (two seeds; dashed: base).

Interpretation. No regulariser helps and nothing was adopted. S1: dropout 0.1-0.3 gives 0.9445-0.9450 (a tie with 0.9441), dropout 0.5 hurts (0.9408), batch-norm and label smoothing are slightly lower, weight decay has no effect on a 1.6k-parameter model, and input noise hurts increasingly (0.9428 -> 0.9399). S2: every setting is within 0.6638-0.6642 except batch-norm (0.6624). The reason is that **these networks do not over-fit** (train accuracy = validation accuracy), so there is nothing for regularisation to repair - the opposite of the face-image task.

3.7 Optimiser, learning rate, batch size and schedule

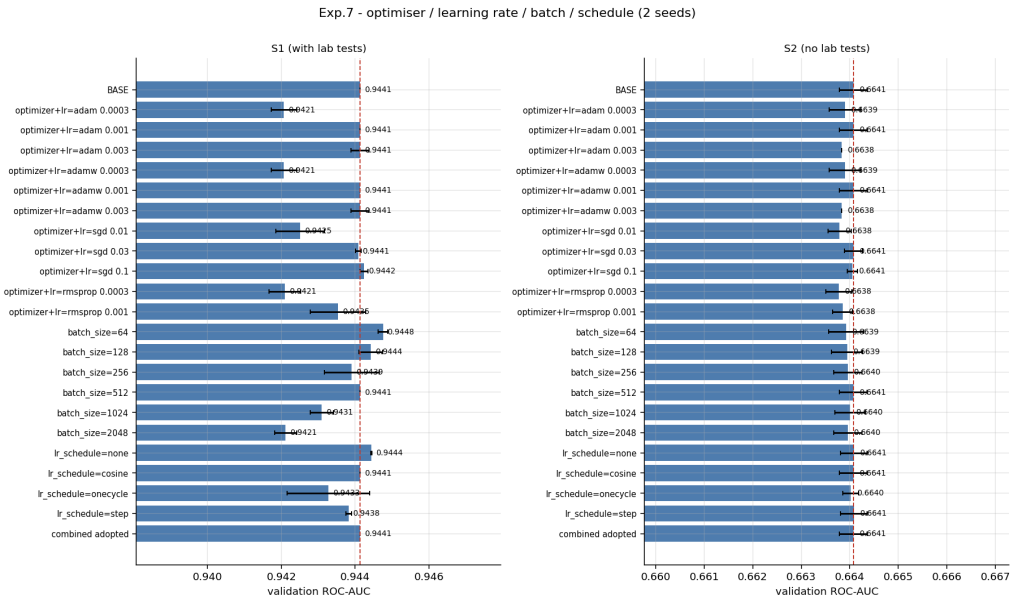


Figure 6. Optimiser, learning rate, batch size and schedule (two seeds; non-AdamW optimisers use weight decay 0).

Interpretation. A learning rate of 3e-4 is too small for 40 epochs (0.9421); 1e-3 and 3e-3 are equal; SGD with Nesterov momentum works with a large rate (0.1: 0.9442). **Smaller batches are slightly better** in S1 (64: 0.9448, 512: 0.9441, 2048: 0.9421) because large batches give too few updates on a small problem - the

opposite of the face task. Schedules are tied. In S2 everything lies within 0.6638-0.6641. No change exceeded the adoption threshold, so the base configuration is kept.

3.8 Loss function and class balance

Plain binary cross-entropy equals focal loss in AUC (S1 0.9441 vs 0.9442-0.9443; S2 0.6641). **Class weighting lowers accuracy** (S1 91.3 %, S2 60.6 %) and AUC slightly, because it moves the decision point away from the true 60 / 40 prior; with such a mild imbalance it solves no problem, and changing the decision threshold (Section 4) is the better tool. Not adopted.

3.9 Joint random search and final-model selection

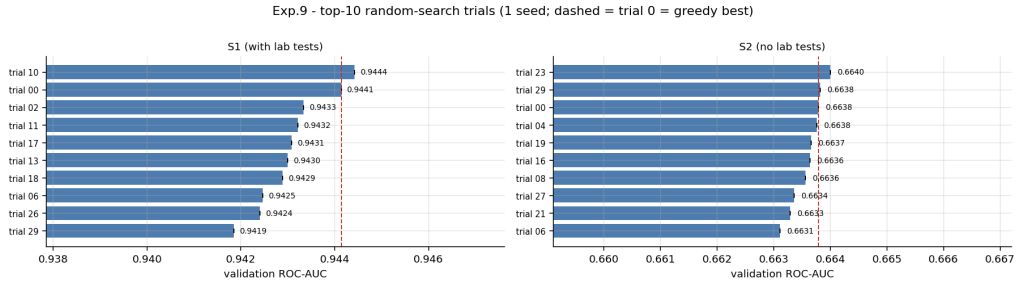


Figure 7. Top-10 random-search trials per scenario (one seed; dashed: the greedy configuration, trial 0).

Thirty random configurations per scenario (architecture, activation, dropout, batch-norm, optimiser, learning rate, weight decay, batch size, label smoothing, scaler, engineered features) span 0.9371-0.9444 in S1 and 0.6586-0.6640 in S2; the greedy configuration (0.9441 / 0.6638) is already within 0.0003 of the best random trial, so a joint search finds nothing better. Re-training the five best configurations seen anywhere with three fresh seeds gives 0.9434-0.9447 (S1) and 0.6640-0.6641 (S2): they are **statistically tied**. The final model is therefore chosen among finalists within 0.0005 AUC by the **lowest validation log-loss** (best-calibrated probabilities). *Process note:* this tie-break was added after a first complete run in which an AUC-only choice had picked a focal-loss model with distorted probabilities (calibration error 0.072 instead of 0.012); a second flaw found in that first run - a 0-hidden-layer model being selectable in S2 - was also corrected before the final run.

Cross-check with scikit-learn. An independent `MLPClassifier` with the same architecture reaches validation AUC 0.944 / accuracy 91.6 % in S1 and AUC 0.659 / accuracy 63.3 % in S2 - within 0.005 AUC of our PyTorch networks (0.944 / 0.664), which supports the correctness of the training code.

3.10 Progress of the tuning

Stage	S1 AUC	S1 acc.	S2 AUC	S2 acc.
1 baseline	0.9421	0.9077	0.6624	0.6332
2 + preprocessing	0.9421	0.9079	0.6621	0.6330
3 + architecture	0.9441	0.9181	0.6637	0.6347
4 + activation	0.9441	0.9181	0.6641	0.6336
5 + regularisation	0.9441	0.9181	0.6641	0.6336
6 + optimisation	0.9441	0.9181	0.6641	0.6336
7 + loss	0.9441	0.9181	0.6641	0.6336
8 final (re-checked, 3 fresh seeds)	0.9445	0.9183	0.6640	0.6330

The only real gains come from using a hidden layer in S1 (+0.002 AUC / +1.0 point accuracy over the 128-64 baseline, partly through the smaller 4 x 16 shape); everything else is within noise. This flat progress is itself the main result: the model is not the limiting factor.

4. Final models and test results

Each final model is a 4 x 16 MLP: **S1** ReLU, dropout 0.2, Adam (lr 1e-3), batch 512, standard scaling, one-hot categoricals (1,585 parameters); **S2** ELU, AdamW (lr 1e-3, weight decay 0.05), same preprocessing (1,521 parameters). Each was trained with five new seeds and evaluated **once** on the 15,000 test patients; the "ensemble" averages the five probabilities.

Model (test set)	Accuracy	AUC	F1	Prec.	Sens.	Spec.	Brier	ECE
S1 ensemble @0.5	0.9203	0.9442	0.9289	0.998	0.869	0.998	0.0663	0.0068
S1 single MLPs (5 seeds)	0.9188 +- 0.0008	0.9436	0.9277					
S1 reference: rule HbA1c >= 6.5	0.9104	0.9253	0.9193	1.000	0.851	1.000		
S2 ensemble @0.5	0.6279	0.6522	0.7232	0.653	0.810	0.354	0.2235	0.0120
S2 single MLPs (5 seeds)	0.6273 +- 0.0014	0.6518	0.7228					
Reference: always "diabetic"	0.5999	0.5000						

Bootstrap 95 % confidence intervals of the ensemble: S1 accuracy 0.9161-0.9245, AUC 0.9406-0.9480; S2 accuracy 0.6201-0.6354, AUC 0.6433-0.6602. ECE = expected calibration error. The S2 interval excludes the majority-class accuracy (0.600), so even this weak model is genuinely better than guessing.

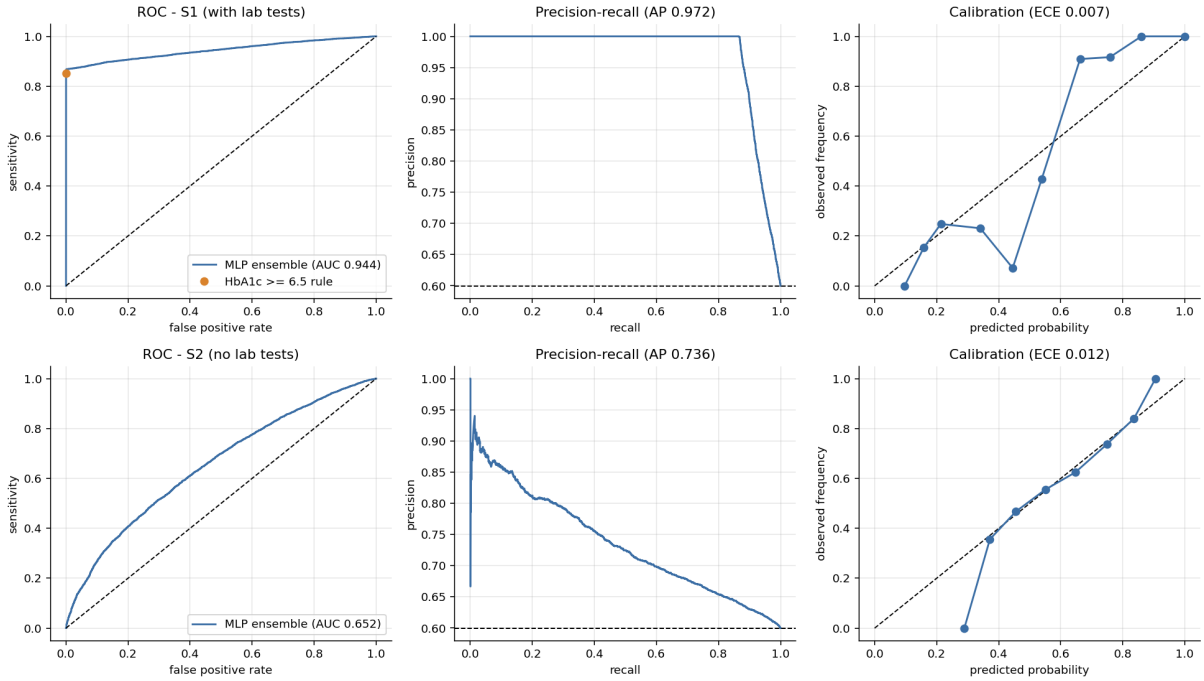


Figure 8. Test ROC, precision-recall and calibration curves (top: S1, bottom: S2). The orange dot is the HbA1c >= 6.5 rule. The S1 curves are step-like because the probabilities are nearly 0 or 1 for most patients (so the middle of the calibration curve is noisy); S2 is well calibrated but barely above the diagonal.

Decision threshold. Thresholds were chosen on the validation set only. A risk model is used with a threshold that fits the purpose:

Setting	Threshold	Accuracy	Precision	Sensitivity	Specificity	F1
S1 default	0.50	0.920	0.998	0.869	0.998	0.929
S1 max validation F1	0.59	0.920	0.999	0.868	0.999	0.929
S1 sensitivity >= 0.90	0.20	0.875	0.892	0.901	0.836	0.897
S2 default	0.50	0.628	0.653	0.810	0.354	0.723
S2 max validation F1	0.37	0.609	0.607	0.987	0.041	0.752
S2 sensitivity >= 0.90	0.45	0.623	0.631	0.897	0.214	0.741

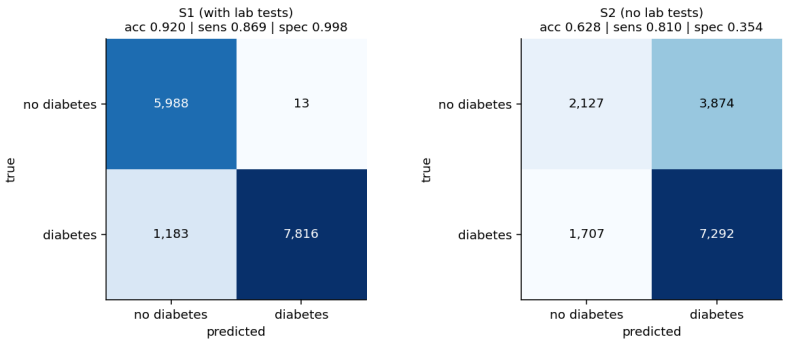


Figure 9. Test confusion matrices (threshold 0.5). S1: almost no false positives, 13 % of the diabetics missed; S2: most diabetics are flagged (sensitivity 0.81) but only a third of the healthy patients are cleared.

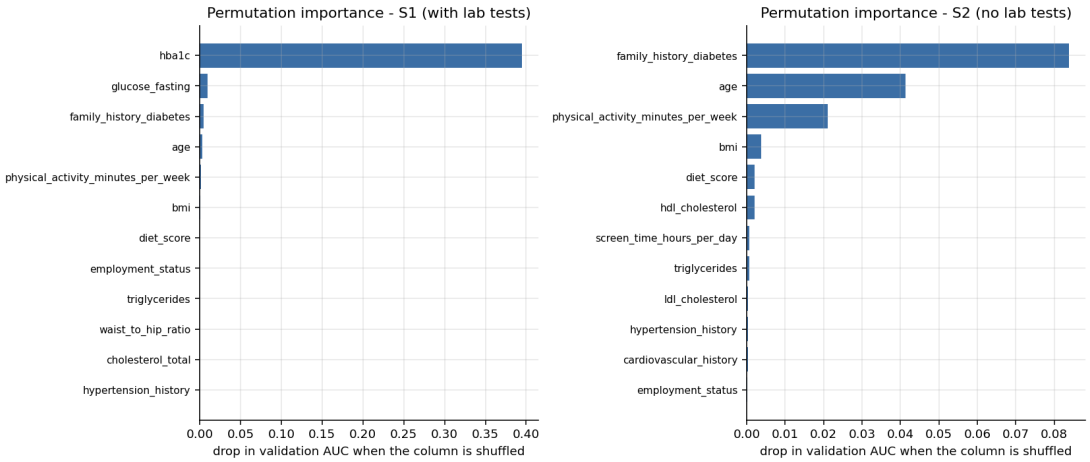


Figure 10. Permutation importance on the validation set: the drop in AUC when one column is shuffled.

Interpretation. S1: shuffling HbA1c lowers AUC by 0.39; the next column, fasting glucose, lowers it by only 0.010. The MLP is, to a first approximation, a smooth version of the clinical threshold rule HbA1c >= 6.5 (rule: sensitivity 0.851, specificity 1.000; MLP: 0.869 / 0.998) with a gain of about one point of accuracy from glucose and risk factors. Its errors are almost all *missed* diabetics: roughly 13 % of the positives lie below the HbA1c threshold and could not be identified from any column, with any architecture, which suggests (but does not prove) that part of the synthetic label is unrelated to the recorded measurements. **S2:** the informative columns are family history (0.084), age (0.041) and physical activity (0.021); everything else contributes less than 0.005. The model ranks patients slightly better than chance (AUC 0.65) and its probabilities are honest (ECE 0.012), but at threshold 0.5 it flags nearly everybody and is useful only to prioritise screening, not to diagnose.

Subgroup performance

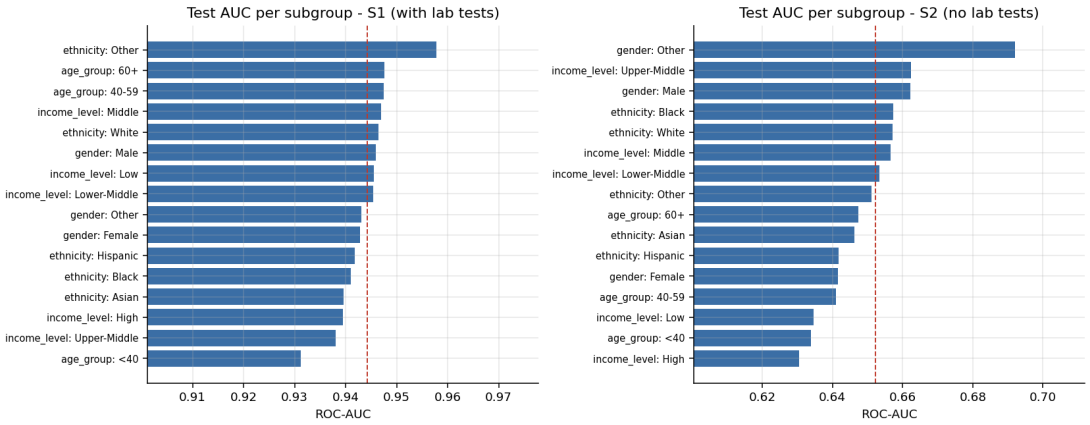


Figure 11. Test AUC per subgroup (dashed: overall).

S1 AUC ranges from 0.931 (patients under 40, sensitivity 0.83) to 0.958 (ethnicity "Other", only 717 patients); S2 AUC ranges from 0.630 (high income, 772 patients) to 0.692 (gender "Other", 276 patients). The

differences are small and the smallest groups are noisy, so we see no evidence that a subgroup is badly served - but a sample of this size cannot rule out small gaps. S2 behaves very differently by age at threshold 0.5 (patients over 60 are almost all flagged, sensitivity 0.97 and specificity 0.06) because age is one of its main inputs and prevalence rises with age (67 % vs 53 % under 40).

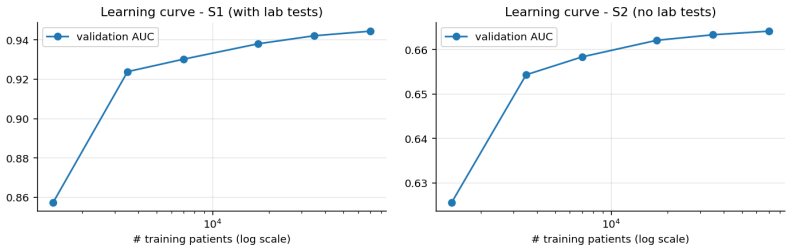


Figure 12. Learning curves of the final configurations. S1 rises from 0.857 (1,400 patients) to 0.944 and gains only +0.002 AUC from 35k to 70k patients; S2 gains +0.002 from 17.5k to 70k.

5. Discussion

Why is tuning so unimportant here? In the face task the models memorised the training set and every regulariser or architecture mattered a little. Here the training and validation accuracies are equal, tiny networks are as good as large ones, and regularisation, optimiser and loss changes move the AUC by less than 0.002. The models are *information-limited*: S1 is determined by one measurement (HbA1c) plus a little glucose information, and S2 contains little signal. Capacity cannot create information that the columns do not contain; the ceiling for S1 is about AUC 0.944 and for S2 about 0.66.

What does the MLP add? Compared with the clinical rule, the MLP gives calibrated probabilities (a continuous risk), a somewhat better sensitivity (0.869 vs 0.851) at almost the same specificity, and the freedom to choose a threshold (e.g. sensitivity 0.90 at specificity 0.84). Compared with a linear model it gains six points of accuracy in S1 (91.8 % vs 85.4 %) because the relation to HbA1c is a threshold, not a line. In S2 it adds nothing over a linear model.

Threats to validity and limitations.

- **Synthetic data:** the relationships come from a generator, not from patients; the "rediscovery" of the HbA1c rule probably reflects how the diagnosis was simulated. Do not use these models for medical decisions.
- **Leakage control:** `diabetes_stage` was excluded and demonstrated to leak; `diabetes_risk_score` is redundant. Results that include them (e.g. 99.9 % accuracy) are not risk predictions.
- **Noise-level comparisons:** most sweep differences are 0.001-0.002 AUC, comparable to the seed-to-seed spread; we did not claim improvements below the adoption threshold, and the final model was chosen among statistical ties by calibration (a rule added after seeing a first complete run, as described in Section 3.9).
- **One split and 2-5 seeds** per configuration; the test CIs only reflect sampling of the 15,000 test patients, not the choice of split.
- **Reproducibility:** the final notebook execution re-used saved results of runs whose configuration and seed were identical to earlier executions (checkpointing); GPU kernels are not bit-wise deterministic, so a fresh run can differ in the 4th decimal.

6. Conclusion

With MLPs only, diabetes can be detected with **92.0 % accuracy (AUC 0.944)** when lab values are available - essentially by learning the HbA1c threshold - and only weakly (**62.8 % accuracy, AUC 0.652**) from demographics, lifestyle and vitals. Careful leakage control (excluding the diabetes stage), a fair protocol (validation for decisions, test once), calibration analysis and subgroup checks matter more than model size: the networks are as small as 1.5k parameters and tuning them further does not help. More informative features, not larger networks, would be needed to improve the screening scenario, and real clinical data would be needed before any practical conclusion.

Reproducibility: run diabetes_risk_mlp.ipynb top to bottom (the dataset is fetched with `kagglehub`*); Python 3.11, PyTorch 2.11 (CUDA), scikit-learn 1.7; about 45 minutes on an RTX 4060 laptop GPU. All evaluated configurations are stored in results/all_runs.csv and the final metrics in results/final_results.json.*